MOHAMED SY

Ph.D. candidate, King Abdullah University of Science and Technology (KAUST)

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PROFESSIONAL SUMMARY

Ph.D. candidate in Mechanical Engineering at KAUST with deep expertise in laser-based spectroscopy, optical diagnostics, and AI-enhanced gas sensing. Hands-on experience with UV–MIR lasers, liquid/gas-phase diagnostics, and advanced signal processing. Track record of innovation (7 patents, 20+ publications), with extensive collaboration on R&D projects in liquid and gad sensing systems.

EXPERIENCE

Research Assistant

Clean Combustion Research Center

August 2020 – Ongoing Thuwal, KSA

- My research consists of using advanced AI techniques to advance gas sensing and combustion diagnostics.
- Led multiple industrial projects focused on multi-species detection, leak detection, and localization.
- Led the development of laser-based sensors for combustion applications and environmental monitoring
- Precision spectroscopy: spectroscopic techniques to measure spectral and molecular parameters
- Wavelength analysis: spectral simulations (HITRAN), selectivity and immunity to interference, and sensitivity analysis.
- Nonlinear optics: optical parametric oscillation (OPO) setup.
- Kinetic simulations: conducting combustion simulations using kinetic mechanisms on CHEMKIN-PRO.
- Hands-on experience with chemical reactors, e.g., shock tubes: multi-speciation time-histories.
- Experienced in the strategic design of experiments, leveraging statistical methods to drive data-informed decisions.
- **Lead/Co-authored nine funded research proposals:**
  1. Natural gas speciation using laser-based sensors (Accepted, *Aramco EXPEC-ARC*, Dhahran)
  2. Laser-based sensors for gaseous leak detection in open fields (Accepted, *Aramco EXPEC-ARC*, KAUST)
  3. BTEX sensing for environmental monitoring (Accepted, EPD, Aramco)
  4. Selective and sensitive laser based sensor for cardiovascular diseases biomarkers and early detection (Accepted,TRG)
  5. Flares emissions monitoring and burning efficiency through LAS (Pending, *Aramco EXPEC-ARC*, KAUST)
  6. Laser based analyzer for alkenes and Drill bit metamorphism correction (EXPEC-ARC, Dhahran).
  7. Liquid and gases water isotopes analyzer using laser absorption spectroscopy and ML (EXPEC-ARC, KAUST)
  8. Rapid and sensitive laser based VOCs sensor for early cancer detection (Pending, Saudi NIH)
  9. A methane sensor to asses cows glazing and digestive systems (Pending, Estidama)

- Presented at several international conferences, e.g., CLEO2023, CLEO2024, LACSEA and the ProCI
- Peer-reviewed several publications

- **Teaching and Mentoring**
- Supervised undergraduate interns and assigned tasks and research projects to them
- Teaching assistant in Fluid Mechanics, AMCS 201 (Writing exams, assignments, projects, and providing practice sessions)
- Organizing Mechanical Engineering PhD Qualifying mock exams (Four semesters)

Summer intern

Aramco (EXPEC ARC)

June 2024 – August 2024 Thuwal, KSA

- Developed an AI-aided model for GHG emission detection, leveraging laser absorption spectroscopy and machine learning techniques.
- Created an ML model for the removal of unknown interferences, significantly enhancing GHG emission detection accuracy.

- Filed a patent for the innovative laser-based system and accompanying ML technique.
- Submitted two Q1 journal papers based on the work completed during the internship.

Combustion school intern
Princeton University

June 2023 – July 2023 | New Jersey, USA

- Advanced course on laser absorption, combustion kinetics and combustion diagnostics.

EDUCATION

Ph.D. in Mechanical Engineering
King Abdullah University of Science and Technology (KAUST)

2022 – Present | Thuwal, KSA

- Ph.D. Topic: Machine-learning techniques to advance multi-component gas sensing: Tackling noise, interference and unknown reference spectra.
- Advisor: Aamir Farooq
- Courses: Low resource deep learning, Laser-based diagnostics, Deep learning, Applied Mathematics 201 and 202, Applied Numerical Methods, Finite Elements Methods, Thermodynamics I and II, Fluid mechanics I and II.

M.S. in Mechanical Engineering
King Abdullah University of Science and Technology (KAUST)

2020 – 2022 | Thuwal, KSA

- Advisor: Aamir Farooq
- Courses: Applied Mathematics I and II, Nonlinear PDEs, Numerical Solutions to Partial Differential Equations, Solid Mechanism, Fluid Mechanics A and B, Introduction to Python, Thermodynamics A and B, Finite Elements Methods.

B.Eng. in Mechanical Engineering and Complex Systems
Tunisia Polytechnic School

2017 – 2020 | Tunis, Tunisia

- Dissertation Topic: Laser-based sensing to toxic gases detection: study case BTEX.
- Advisors: Fehmi Najjar and Aamir Farooq
- Grade Point Average: 18.5 out of 20
- Courses: Mathematics for Engineers, Operations Research, Ordinary and Partial Differential Equations, Robotics, Fluid Mechanics, Introduction to C++, Discrete Mathematics, Fundamentals of Economics, Data Structures and Algorithms, Discrete Mathematics, Introduction to Data Science, Fundamentals of Economics, Fundamentals of Mathematics, Linear Algebra I & II, Math Softwares, Mathematical Analysis I & II, Operations Research, Optimization, Stochastic Process, Probability and Statistics I & II, Differential Equations, Marketing, Game Theory, Introduction to Law and Policy.

B.S. in Math and Physics (MP)
Ecole Nationale aux Etudes d'Engernieure de Sfax (IPEIS)

2015 – 2017 | Sfax, Tunisia

- Ranked 51st among 4000 candidates for the national exams to enter engineering schools
- Courses: Algebra I & II, Calculus I & II, Data Structures and Algorithms, Fundamentals of Physics, Organic Chemistry, and Advanced Calculus, Advanced Algebra.

AWARDS

Recipient of the 2024 Physical Science and Engineering Dean's Award for outstanding PhD students
King Abdullah University of Science and Technology

Ranked first
Mechanical engineering program at Tunisia Polytechnic School

Ranked TOP 1%

National exam for entrance to engineering school 51st/4000 students

Recipient of a five years scholarship
Recipient of scholarship of 5 years from Mauritanian government based on merit.

PATENTS AND INNOVATIONS

Method and system for water cut sensing in an oil-water flow
US Patent App. 18/535,964, June 26, 2024.

Unknown spectral interference removal technique
U.S. Patent with Aramco in preparation

Auto-Ranging Dual-Path Gas Sensor for Continuous On-Site Monitoring of Alkanes, CO_2, and H_2S
U.S. Patent with Aramco in preparation

ML enhanced algorithms for Continuous On-Site Monitoring of Alkanes, CO_2, and H_2S
U.S. Patent with Aramco in preparation

Laser-based selective btex sensing with deep neural network
US Patent App. 17/901,068, March 02, 2023.

Flare emission monitoring through LAS
U.S. Patent with Aramco in preparation

PUBLICATIONS

Journal Articles

[1] **Sy, Mohamed**, et al. "Laser-based speciation of isoprene thermal decomposition behind reflected shock waves." *Proceedings of the Combustion Institute* 40.1-4 (2024): 105460.

[2] **Sy, Mohamed**, Mhanna Mhanna, and Aamir Farooq. "Multi-speciation in shock tube experiments using a single laser and deep neural networks." Combustion and Flame 255 (2023): 112929.

[3] **Sy, Mohamed**, Emad Al Ibrahim, Aamir Farooq " VOC-certifire: Certifiably robust one-shot spectroscopic gases classification via randomized smoothing " ACS-OMEGA

[4] **Sy, Mohamed**, Emad Al Ibrahim, Ali Elkhazraji, Aamir Farooq " Unsupervised Source Separation: Quantifying hydrocarbons without their reference spectra using VAEs" (submitted to ICLS)

[5] **Sy, Mohamed**, Sarah Aamir, Aamir Farooq " Multi-component gas sensing via spectral feature engineering" (Sensors and Actuators B: Chemical)

[6] **Sy, Mohamed**, Mohammed Sait, Aamir Farooq "AI-aided interference removal technique for enhanced GHGs emission detection" (In review, IEEE SENSORS)

[7] **Sy, Mohamed** et al. "Reference-Free Multi-Species Gas Detection via Unsupervised Learning" (IEEE SENSORS, 2024)

[8] **Sy, Mohamed**, et al. "ML-enhanced compact, sensitive and selective C1-C5 natural gas analyzer using single laser and a mutlipass cell" (LACSEA)

[9] **Sy, Mohamed**, Emad Al Ibrahim, Aamir Farooq " MSC-certifire: Mixture spectra classifier, a certifiably robust one-shot spectroscopic gases mixture detection via randomized smoothing" ACS-OMEGA

[10] **Sy, Mohamed**, Shams Musheer, Farhath Syed, and Aamir Farooq "Blind Source Separation for Multiple Species Detection: Extension to High Temperatures and High Pressures"(Manuscript in preparation).

[11] **Sy, Mohamed**, Muhammad S. Kham, Ali Elkhazraji, Aamir Farooq "ML- enhanced simultaneous H2S/CO2 and C1-C5 compact gas analyzer using time multiplexing and CNNs for oil and gas applications" (Manuscript in preparation).

[12] **Sy, Mohamed**, Muhammad S. Kham, Ali Elkhazraji, Aamir Farooq "A wide dynamic range natural gas analyzer for Mud lugging and well reservoirs using LAS combined single pass/multipass cell" (Manuscript submitted to ICLS).

[13] **Sy, Mohamed**, Muhammad S. Kham, Ali Elkhazraji, Aamir Farooq "Speciation and kinetics modeling of isoprene and 1,3-Butadiene oxidation behind reflected shock waves" (AIAA Scitech 2024).

[14] Mhanna, Mhanna, **Mohamed, Sy**, Aamir Farooq. "Laser-based selective BTEX sensing using deep neural networks." Optics Letters 47.13 (2022): 3247-3250.

[15] Mhanna, Mhanna, **Mohamed Sy**, Ali Elkhazraji, and Aamir Farooq. "Deep neural networks for simultaneous BTEX sensing at high temperatures." Optics Express 30, no. 21 (2022): 38550-38563.

[16] Mhanna, Mhanna, **Mohamed Sy,** and Aamir Farooq. "A selective laser-based sensor for fugitive methane emissions." *Scientific reports* 13, no. 1 (2023): 1573.

[17] Elkhazraji, Ali, **Mohamed Sy**, Mhanna Mhanna, Joury Aldhawyan, Mohammad Khaled Shakfa, and Aamir Farooq. "Selective BTEX detection using laser absorption spectroscopy in the CH bending mode region." *Experimental Thermal and Fluid Science* 151 (2024): 111090.

[18] Mhanna, Mhanna, **Mohamed Sy**, Ali Elkhazraji, and Aamir Farooq. "A laser-based sensor for selective detection of benzene, acetylene, and carbon dioxide in the fingerprint region." *Applied Physics B* 129, no. 9 (2023): 139.

[19] Elkhazraji, Ali, Mohammad Khaled Shakfa, Nawaf Abualsaud, Mhanna Mhanna, **Mohamed Sy**, Marco Marangoni, and Aamir Farooq. "Laser-based sensing in the long-wavelength mid-infrared: chemical kinetics and environmental monitoring applications." *Applied Optics* 62, no. 6 (2023): A46-A58.

[20] Mhanna, Mhanna, **Mohamed Sy**, Ayman Arfaj, Jose Llamas, and Aamir Farooq. "Highly sensitive and selective laser-based BTEX sensor for occupational and environmental monitoring." *Applied Optics* 63, no. 11 (2024): 2892-2899.

[21] Elkhazraji, Ali, **Mohamed Sy**, Mohammad Khaled Shakfa, and Aamir Farooq. "A mid-infrared laser diagnostic for simultaneous

detection of furan and nitric oxide." *Proceedings of the Combustion Institute* 40, no. 1-4 (2024): 105366.
[22] Jiabiao Zou, Mohammad Adil, Ali Elkhazraji, **Mohamed Sy**, Mhanna Mhanna, Aamir Farooq. “Shock tube study of alkanes low-temperature chemistry: Multi-species time-history during the first stage ignition.” *Combustion and Flames*
[23] Jiabiao Zou, Mohammad Adil, Ali Elkhazraji, **Mohamed Sy**, Mhanna Mhanna, Aamir Farooq. “Shock tube study of cycloalkanes low-temperature chemistry: Multi-species time-history during the first stage ignition” *Combustion and Flames*

RECENT IN-PERSON TALKS

- Asia-Pacific Combustion Symposium (ASPACC 2025), Singapore
- European Combustion Meeting (ECM 2025), Edinburgh, UK.
- Conference on Neural Information Processing Systems (NeurIPS 2024), Vancouver, CA.
- Conference on Lasers and Electro-Optics (CLEO 2024), Charlotte, NC, USA.
- CI’s 40th International Symposium - Emphasizing Energy Transition (PROCI 2024), Milan, Italy.
- Laser Applications to Chemical, Security and Environmental Analysis (LACSEA 2024), Toulouse, France.
- Field Laser Applications in Industry and Research (FLAIR 2024), Assisi, Italy.
- Mediterranean Combustion Symposium (MCS 2023), Luxor, Egypt.
- Conference on Lasers and Electro-Optics (CLEO 2023), San Jose, CA, USA
- 39th International Symposium on Combustion (PROCI 2022), Vancouver, Canada.

SKILLS

- Absorption Spectroscopy, FTIR and Spectroscopy Tools such as HAPI, GRAMS AI, PLS Toolbox.
- **Data Analysis:** Python (NumPy, SciPy, Scikit-learn), Matlab, LabVIEW, Origin, Horizon, IGOR Pro
- **ML/AI Frameworks:** PyTorch, TensorFlow, Keras, Autograd
- **Hardware & Optics:** ICOS, multipass cells, diode lasers, and OPO systems.

LANGUAGES

- English (Fluent)
- French (Fluent)
- Arabic (Fluent)
- Fulani (Native)

REFERENCES

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